



2017



PROJECT SUBMITTAL

D48B7.5X – 4LWC

DRUM FAN WITH FOUR LEGS AND CASTERS

SUBMITTED TO | Alton Steel

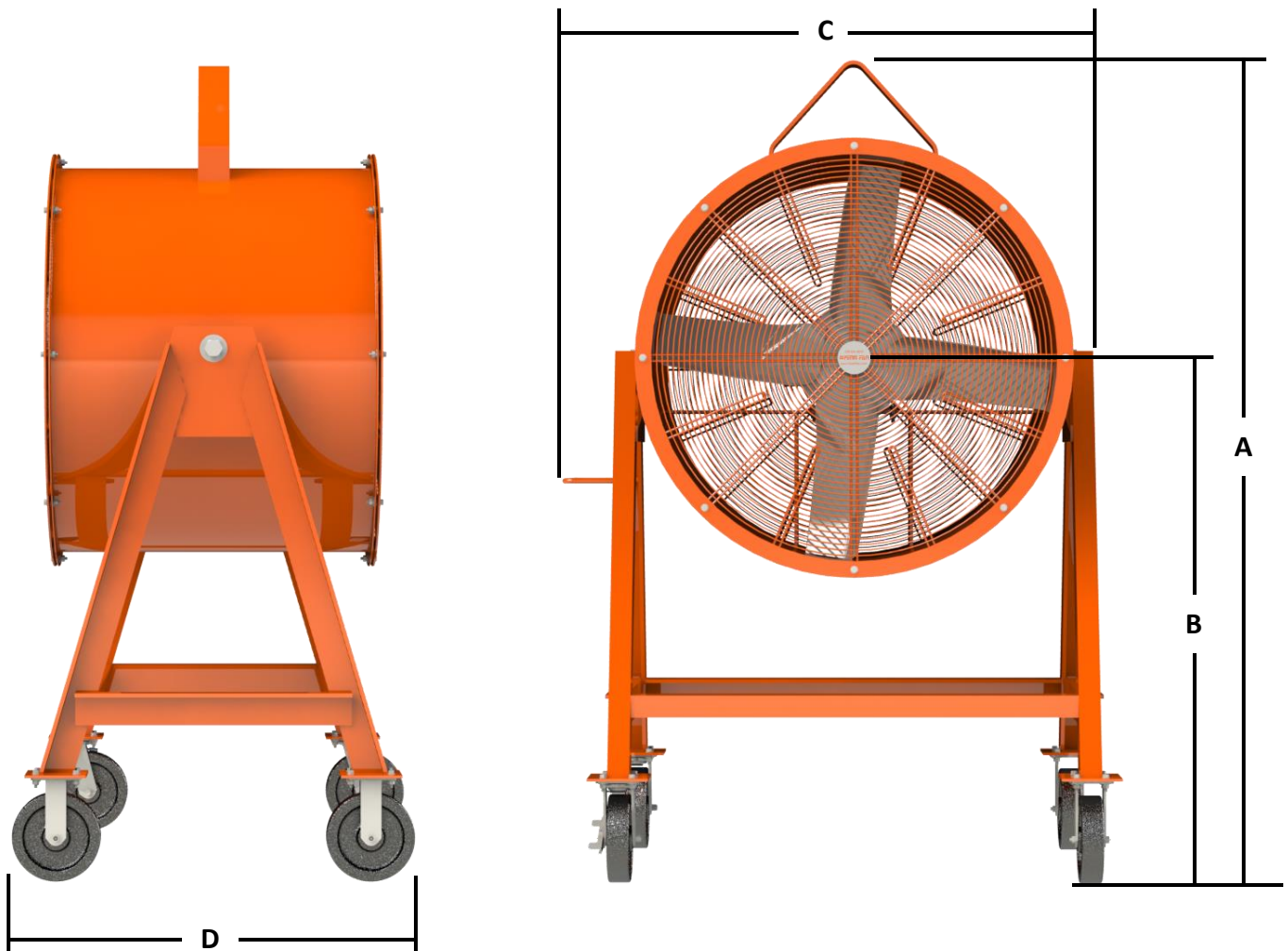


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PRODUCT DESCRIPTION & DIMENSIONS



D48B7.5X – 4LWC Dimensions				
A	B	C	D	Approximate Weight
84.5"	48.5"	60"	34.5"	870lbs

The drum style body offers the same wide range of cooling capabilities as the standard mancooling fan, but with the added feature of greater air direction. This helps condition difficult to reach areas and provides even greater spot cooling of equipment, processes and people. Drum fans provide maximum durability in the harshest of working environments. Penn Fan's Drum fans were designed, engineered and built with industries such as steel mills, forges, foundries and melt shops in mind. Adding the legs and casters provides a stable, easily transferrable unit that can be utilized throughout your facility.

22329 Perry Highway | PO Box 52 | Zelienople, PA 16063
(P) 724-452-4570 | www.PennFan.com | (F) 724-452-0961



Fan Curve Data

Propeller Designation: **X-48-6 XHD**

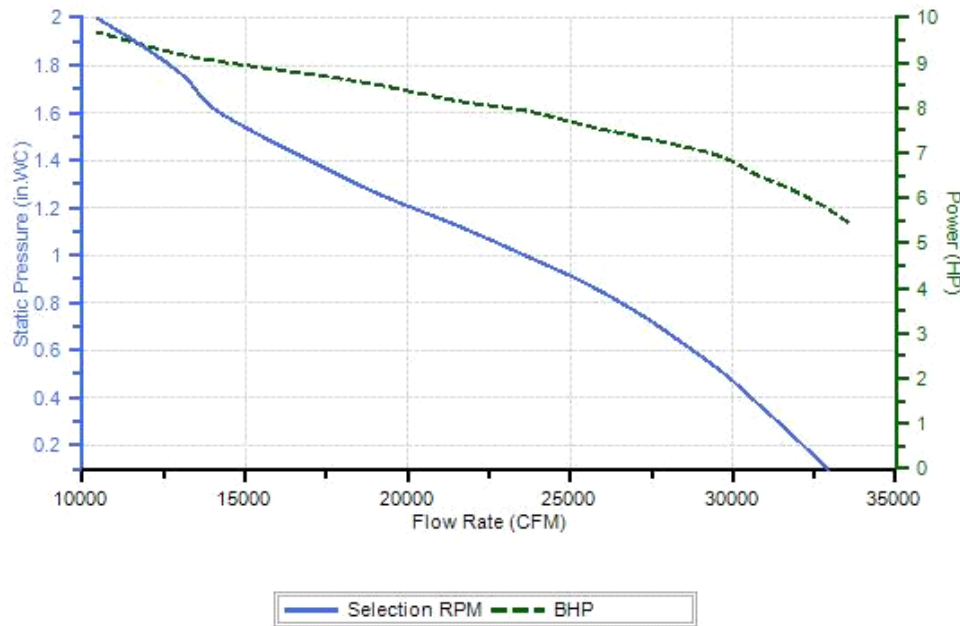
Tag: M48B7.5X

Fan Info

Fan Speed: 1175 RPM
Fan Diameter: 48 in
Air Density: 0.07535 lbm/cu.ft

Construction Limits

	SD	HD	XHD	
Power	3.5	6	10.4	Hp
Speed	960	1140	1360	RPM

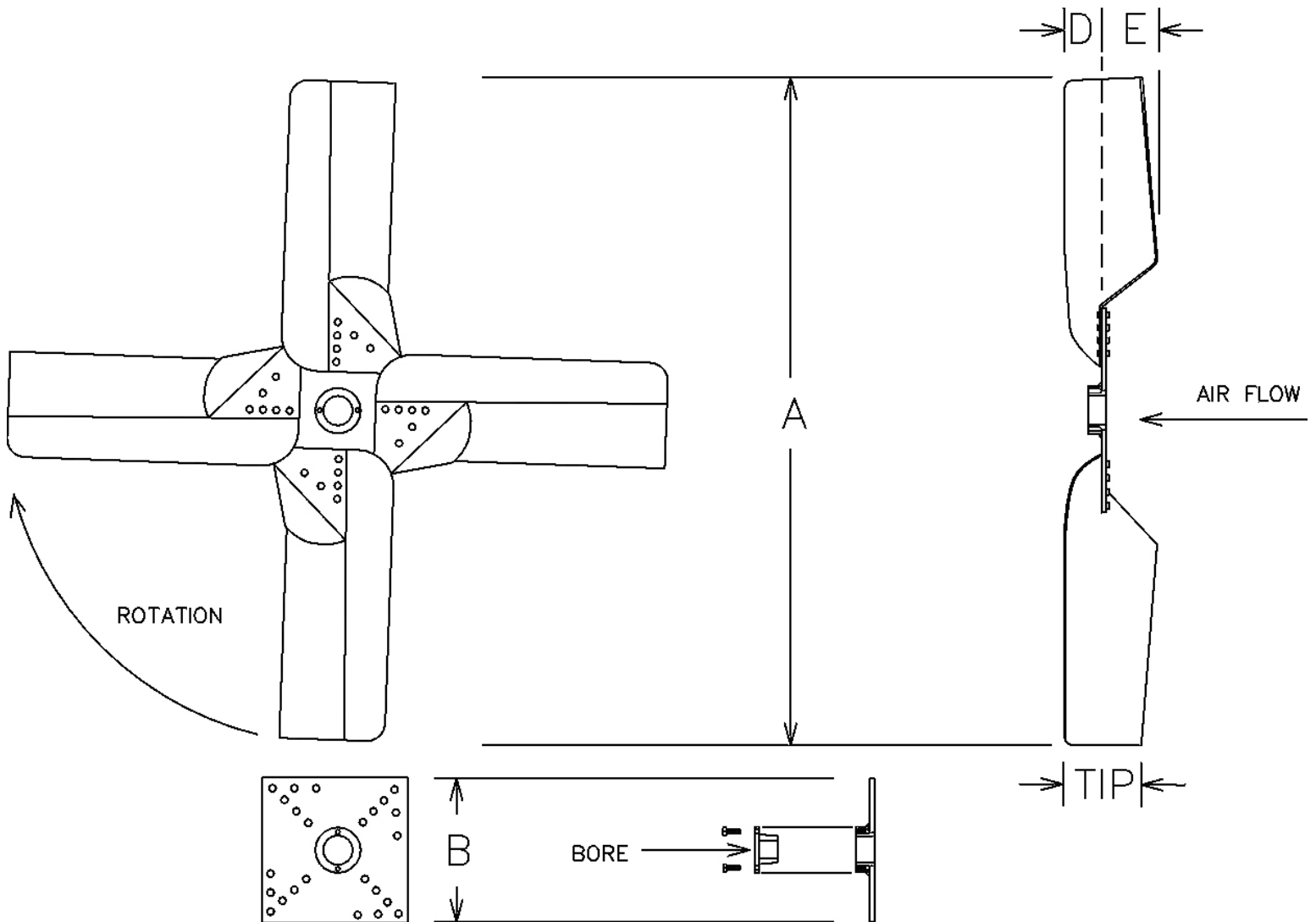


Cnst.	Static Pressure (in.wg)	Flow Rate (CFM)	Power (Hp)	SEFF (%)	TEFF (%)
XHD	0	33563	5.447	0	40
XHD	0.125	32707	5.857	11	45
XHD	0.25	31767	6.209	20	50
XHD	0.375	30751	6.507	28	54
XHD	0.5	29734	6.897	34	56
XHD	0.625	28494	7.145	39	58
XHD	0.75	27161	7.35	44	59
XHD	0.875	25560	7.595	46	59
XHD	1	23591	7.944	47	56
XHD	1.125	21515	8.155	47	54
XHD	1.25	19231	8.494	45	49

All selections are based on CARBON STEEL construction. Test data is accumulated based on AMCA 210.85, figure 12 tests, 2% tip clearance and ANSI S12.12 panel tests inlet only.



PROPELLER DRAWING



EXHAUST - CLOCKWISE

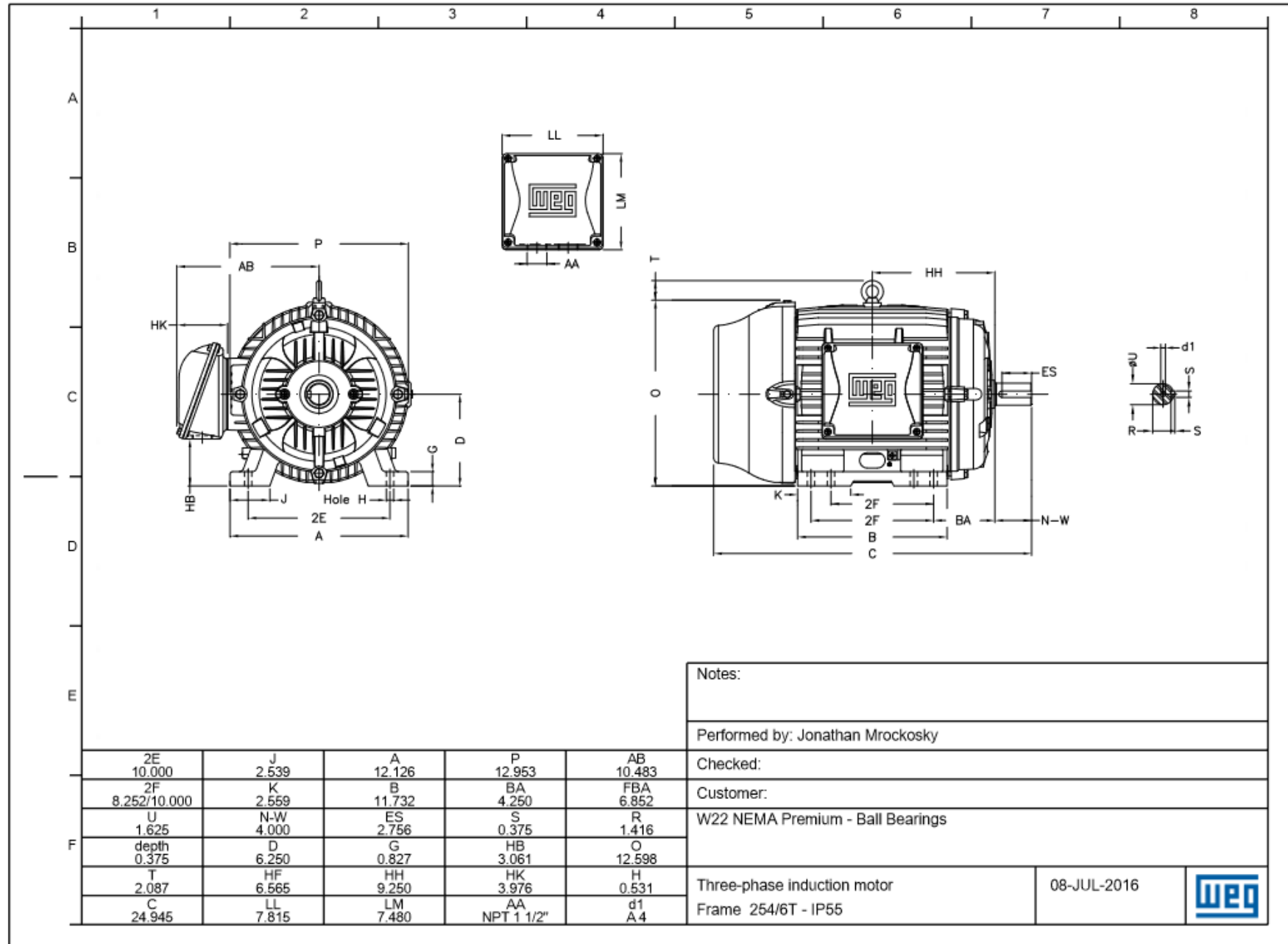
Tolerances are +/- 0.125 inch unless specified

Blade: 10 ga

Hub: .3125 ga

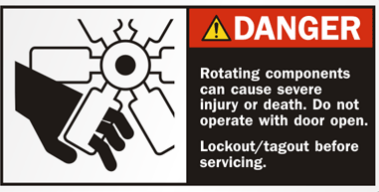
Fastener: 1/4 Rivets

Designation	A	B	D	E	Tip	BSH	Bore	Material	Finish	App Wt
X-48-6 XHD	48	12	1 1/2	1	2 1/8	P	1 5/8	Mill Galvanized Steel	None	38.7661 lb





SAFETY WARNINGS

	DANGER! RISK OF ELECTRIC SHOCK OR ELECTROCUTION! DANGEROUS ROTATING EQUIPMENT DANGER OF DEATH, BODILY HARM OR DAMAGE TO PROPERTY!
	Consult a licensed or certified electrician in accordance with the National Electrical Code and all applicable local codes and ordinances. Improper installation will create an electrical hazard, which could result in death or serious injury users, installers or others due to electrical shock, and may also cause damage to property. Always disconnect the power before installing or servicing the unit. Failure to do so could result in death or serious injury to servicemen, users or others due to electrical shock.
	Consult a qualified rigging contractor for lifting and placing fan for installation (not all fans require a rigging and or installation contractor). Some fans come equipped with safety cables. Where these cables are present, they must be used and secured to a structure rated to carry the weight of the unit in the event the primary fastening system should fail. Penn Fan is not responsible for recommending, designing, engineering fastening systems. See a qualified and licensed engineer for assistance.
	Follow all LOCKOUT & TAGOUT procedures. For more information on LOCKOUT & TAGOUT procedures, refer to the OSHA Standard 1910.147. The Standard is available at www.OSHA.gov. Failure to comply with these safety procedures may result in damage to property, injury to persons or death.
	Danger, rotating equipment. DO NOT OPERATE without safety guards in place. Where safety guards are not required, operate with caution ensuring that persons and property are free of the fan when energized. Be cautious of loose clothing or any other item that could be drawn into the fan during operation resulting in injury or death.
	READ and SAVE this manual for future reference. READ the entire document thoroughly and carefully before attempting to assemble, install, operate or perform maintenance on the product. Failure to comply with instructions could result in personal injury and/or property damage.



LIMITED WARRANTEE	Penn Fan Company, the manufacturer, warrants that for a period of 1 year from the date of purchase it will repair, or at its option replace, any new fan which proves defective in material or workmanship. All repairs and replacements shall be F.O.B. factory. All claims must be made in writing and approved by the manufacturer. In no event and under no circumstances shall the manufacturer be liable for (a) damages in shipment; (b) failure due to operation, intentional or otherwise, and (c) non-authorized expenses for removal, inspection, transportation, repair or rework. Nor shall manufacturer ever be liable for consequential and incidental damages, or in any amount greater than the purchase price of the product. This warranty is in LIEU OF ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, ANY IMPLIED WARRANTIES OR MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. THE TERMS OF THIS WARRANTY CONSTITUTE ALL BUYER'S OR USER'S SOLE AND EXCLUSIVE REMEDY, AND ARE IN LIEU OF ANY RIGHTS TO RECOVER FOR NEGLIGENCE, BREACH OF WARRANTY, STRICT TORT LIABILITY OR UPON ANY OTHER THEORY. Any legal proceedings arising out of the sale or use of this product must be commenced within 18 months of the date of purchase.
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Upon receiving your Penn Fan unit, be sure that there are no visible signs of damage before signing for the unit. If you see damage, note the damage on the Bill of Lading and keep a copy. Report the damage to the appropriate department with the freight carrier. Failure to do so may result in the inability to receive compensation for the damaged product.

Lifting or Handling the fan by physical or mechanical means should be done by someone in good physical condition and trained to operate lifting equipment. Ensure that the equipment used is rated for the product delivered. Padded/coated chains, lifting cables or straps and the use of a spreader bar are recommended when needed. Contact a qualified rigging company if you are unsure about lifting or moving the product.

When uncrating or unpacking your Penn Fan unit, be sure to avoid damaging the product. Using pry bars as a method to disable a crate is not recommended. Keep all shipping materials until you have verified the unit does not need to be returned for any reason. Strapping may be under high tension and caution should be taken when cutting these straps.

Penn Fan fabricates a high-quality industrial duty fan. Each Penn Fan unit is tested before it leaves the factory. It is still advisable to run test your fan to ensure proper operation as damage may have occurred during shipping. Fans that are to be installed in an area that will be difficult to access should be tested before installation.

Before run testing or installing your fan, be sure to check the unit for any signs of damage that may have occurred during transportation. As our fans are very large and often shipped in wooden crates, strapped to a pallet, or in large heavy-duty cardboard boxes some minor cosmetic damage is to be expected.

If you see obvious signs of damage to the fan blade(s), STOP. DO NOT run test the fan. If you suspect the fan blade is loose, STOP. DO NOT run test the fan. Please contact your authorized Penn Fan representative or contact the factory.

Penn Fan Company
22329 Perry Highway
Zelienople, PA 16063
(724) 452-4570 Main Line

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